



NATIONAL JUNIOR COLLEGE
SENIOR HIGH 2 PRELIMINARY EXAMINATION
Higher 2

NAME

--

SUBJECT
CLASS

2ma2

REGISTRATION
NUMBER

--	--	--	--	--	--	--

MATHEMATICS

9758/01

Paper 1

15 September 2020

3 hours

Candidates answer on the Question Paper.

Additional Materials: List of Formulae (MF26)
Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, class and registration number on the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use paper clips, glue or correction fluid.

Answer **all** the questions.

Write your answers in the spaces provided in the Question Paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

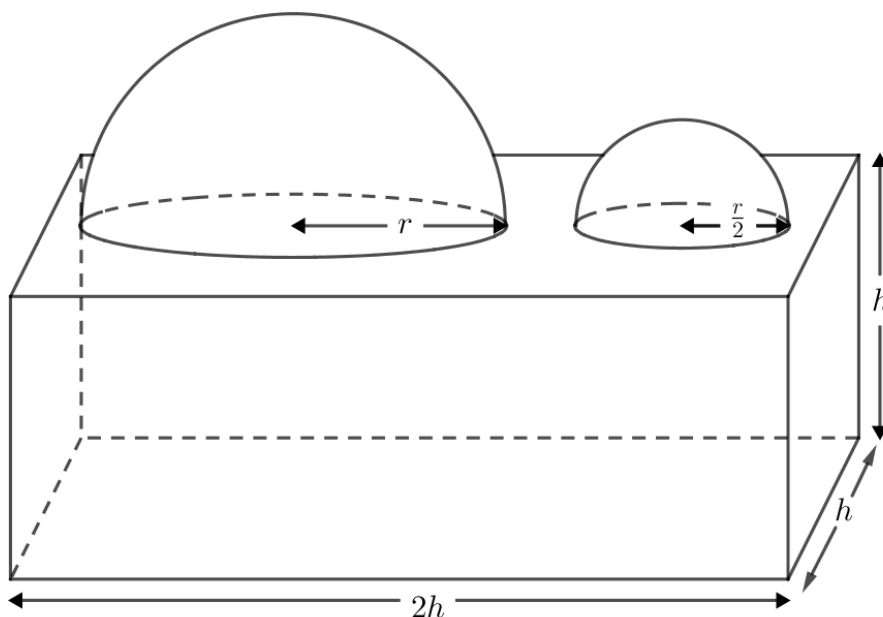
You are reminded of the need for clear presentation in your answers. Up to 2 marks may be deducted for improper presentation.

The number of marks is given in brackets [] at the end of each question or part question.

Question Number	Marks Possible	Marks Obtained
1	4	
2	5	
3	7	
4	8	
5	8	
6	8	
7	8	
8	8	
9	10	
10	10	
11	10	
12	14	
Presentation Deduction		– 1 / – 2
TOTAL	100	

This document consists of 8 printed pages.

- 1 An arithmetic series has first term a and common difference d , where a and d are non-zero. It is given that the sixth, fourth and third terms of the arithmetic series are equal to the first three terms respectively of a convergent geometric series. Find, in terms of a , the sum of the terms of the geometric series after, and including, the 15th term. [4]
- 2 By rationalising $u_r = \frac{1}{\sqrt{r} + \sqrt{r-1}}$, find an expression for $\sum_{r=1}^n u_r$.
Deduce the sum from the seventeenth to the hundredth terms. [5]
- 3 [It is given that a sphere of radius r has surface area $4\pi r^2$ and volume $\frac{4}{3}\pi r^3$.]



A model of an observatory is made up of two parts.

- The main building is modelled by a closed rectangular box with two circular discs of radii r cm and $\frac{r}{2}$ cm removed from the top. The box is of length $2h$ cm, width h cm and height h cm.
- The telescope domes are modelled by the curved surfaces of two hemispheres of radii r cm and $\frac{r}{2}$ cm.

Both parts are joined together as shown in the diagram and the model is made of material of negligible thickness. It is given that the volume of the model is a fixed value 500 cm^3 and the model has external surface area $S \text{ cm}^2$.

Use differentiation to find the value of r that gives a stationary value of S .

Hence, determine whether the value of r found gives a minimum or maximum value of S . [7]

4 It is given that $f(x) = 10 - kx - \frac{9}{x-1}$, where k is a constant.

- (a) In the case where $1 < k < 9$, sketch the graph of $y = f'(x)$, labelling the x -intercepts and the equations of any asymptotes. Leave your answers in terms of k . [3]

For the remainder of the question, assume that $k = 1$.

- (b) Sketch the graph of $y = f(x)$, labelling the equations of any asymptotes. Hence solve the inequality

$$f(x) \geq \frac{9}{|x-1|}. \quad [5]$$

5 The curve C has equation $y = a(cx - b)^2$, where a , b and c are positive constants.

- (i) Describe fully a sequence of (no more than three) transformations which would transform the graph of $y = x^2$ onto the graph of C . [3]
- (ii) Given that C passes through the points with coordinates $(-1, 147)$, $(2, 3)$ and $(4, 27)$, find a and c in terms of b . [3]
- (iii) It is given further that the point with coordinates $(-5, 25)$ on $y = x^2$ corresponds to the y -intercept on $y = a(cx - b)^2$. State the integer values of a , b and c . [2]

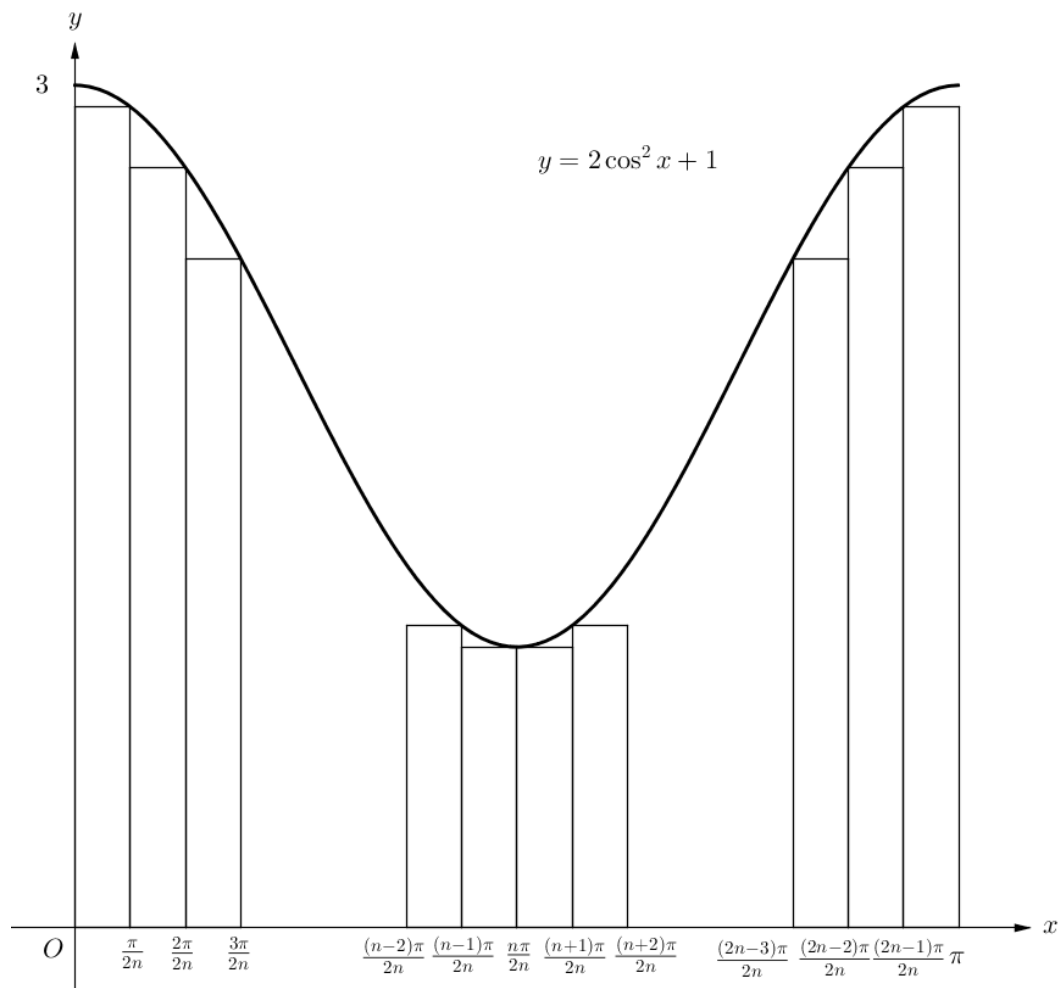
6 The integral test for convergence, known as the Maclaurin-Cauchy test, is a method used to test for convergence of an infinite series of non-negative terms. The Maclaurin-Cauchy test states that

“For a non-negative and non-increasing function f defined on the interval $[1, \infty)$, a series of the form $\sum_{r=1}^{\infty} f(r)$ converges if $\int_1^{\infty} f(x) \, dx$ is finite.”

- (i) Show that $\frac{\tan^{-1} x}{1+x^2}$ is positive and decreasing on the interval $[1, \infty)$. [3]
- (ii) Use the Maclaurin-Cauchy test to determine if $\sum_{r=1}^{\infty} \frac{\tan^{-1} r}{1+r^2}$ converges. [5]

7 (a) Evaluate exactly $\int_0^{\pi} \cos\left(\frac{x}{2}\right)\cos\left(\frac{x}{3}\right) + \pi \, dx$. [2]

(b) (i)



The curve with equation $y = 2 \cos^2 x + 1$, for $0 \leq x \leq \pi$, is shown in the diagram. Rectangles, each of width $\frac{\pi}{2n}$, are drawn under the curve. Show that the total area S_n of all $2n$ rectangles is given by

$$\frac{2\pi}{n} \left\{ \cos^2\left(\frac{\pi}{2n}\right) + \cos^2\left(\frac{2\pi}{2n}\right) + \cdots + \cos^2\left(\frac{(n-1)\pi}{2n}\right) \right\} + \pi. \quad [3]$$

(ii) Find the exact sum to infinity S_{∞} . [3]

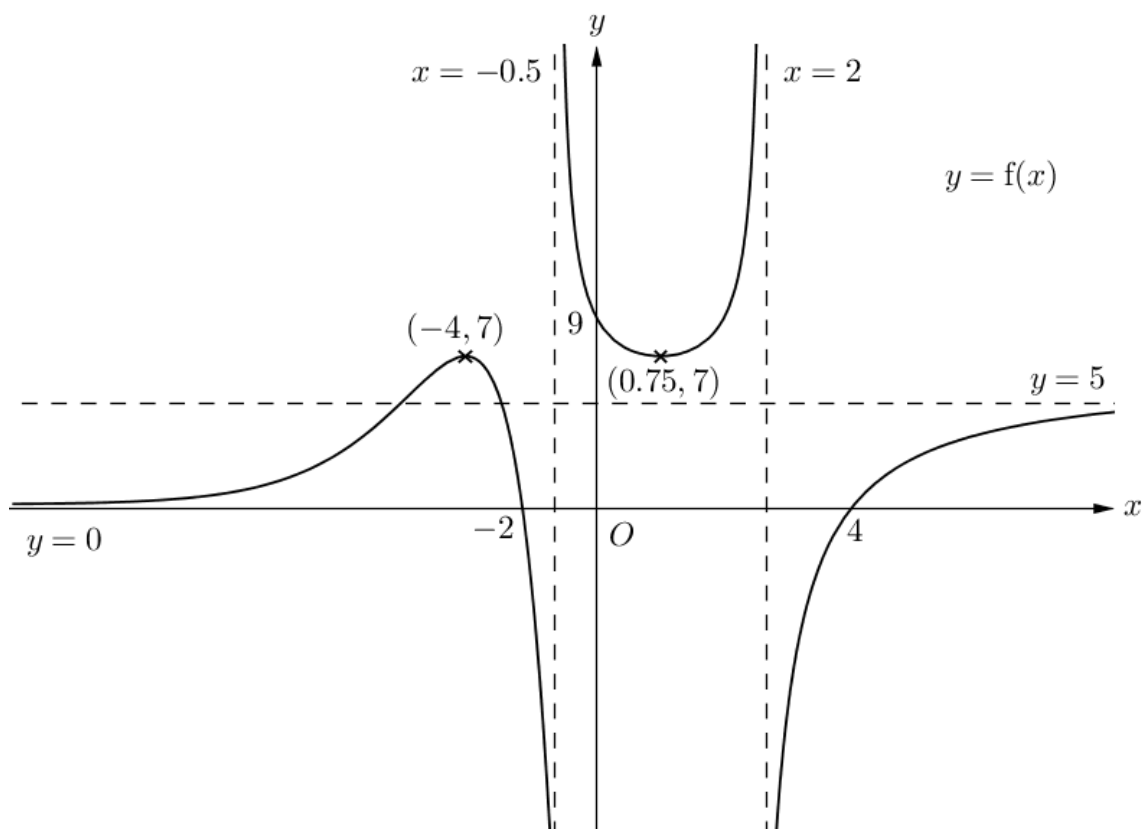
- 8 A blogger reviews a data plan with unlimited data. The download speed is x megabytes per second, t seconds after the download has started. This download speed decreases at a rate proportional to the difference between x and the eventual limit of 40 megabytes per second. The download speed falls from a value of 60 megabytes per second to 50 megabytes per second in the first 30 minutes.

- (i) Write down a differential equation for this situation. Solve this differential equation to get x as an exact function of t . [5]

The total amount of data downloaded daily by time t seconds is y gigabytes. The fair usage policy enforces that the download speed will be reduced to a speed which cannot be used to stream videos if $y > 500$. You may assume that 1 gigabyte is 1000 megabytes.

- (ii) The blogger starts using the data plan on the morning of a particular day to stream videos. How long would he be able to stream videos continuously? Give your answer in hours to 1 decimal place. [3]

9



The diagram shows the curve with equation $y = f(x)$, for $x \in \mathbb{R}$, $x \neq -\frac{1}{2}$, $x \neq 2$. The curve crosses the axes at $x = -2$, $x = 4$ and $y = 9$, and has turning points with coordinates $(-4, 7)$ and $(0.75, 7)$. The curve has asymptotes with equations $x = -0.5$, $x = 2$, $y = 0$ and $y = 5$.

- (i) Sketch the curve $y = \frac{1}{f(x)}$, labelling any axial intercepts and turning points, and the equations of any asymptotes. [4]

For the remainder of the question, the domain of f is $[k, 2) \cup (2, \infty)$, where k is a constant.

- (ii) State the least value of k for which the function f^{-1} exists. [1]

The function g is defined by

$$g : x \mapsto \frac{2x+a}{x-b} \text{ for } x \in \mathbb{R}, x > b,$$

where $a, b \in \mathbb{R}^+$.

- (iii) Find the range of g . [2]

- (iv) Find $(fg)^{-1}(0)$. Leave your answer in terms of a and b . [3]

- 10 A curve C has equation $x^2 = \frac{\ln(y+1)}{(y+1)^2}$. The points P and Q on C each have y -coordinate $e-1$.

- (i) State the equation of the line of symmetry of C . [1]

- (ii) (a) A vase is formed by rotating the region bounded by the curve C and the line PQ through π radians about the y -axis. Find the exact volume of the vase obtained. [4]

- (b) The normals to C at P and Q meet at the point N . Find the exact coordinates of N . [5]

11 Do not use a calculator in answering this question.

- (a) The complex numbers z and w are such that $w - 6\alpha z = -5\alpha i + 3$ and $\alpha z^2 + iw = 10\alpha + 3i$, where α is a non-zero real constant.

Given that $|z| > 1$, find z and w in terms of α (where applicable). [3]

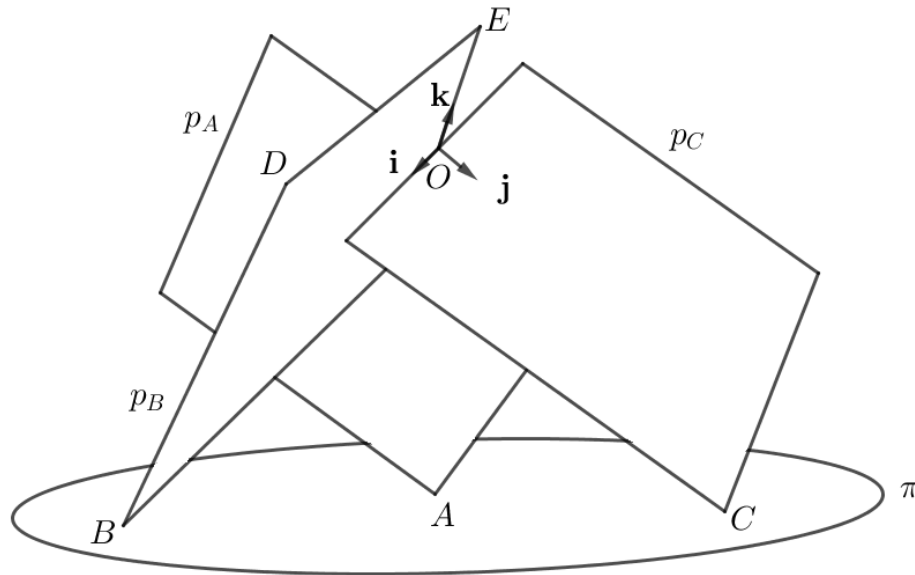
- (b) The complex numbers z_1 and z_2 are given by $3 - i\sqrt{3}$ and $2i$ respectively.

- (i) Find the possible integer values of n such that $\frac{z_1^{3n}}{z_2^{2n}}$ is purely imaginary with a positive imaginary part. [3]

- (ii) Given that z_1 is a root of the equation $px^3 - 9x^2 + 6x + 36 = 0$, find the value of the real number p . Using the value of p found, express $px^3 - 9x^2 + 6x + 36$ as a product of three linear factors. [4]



The picture shows a concrete sculpture in Rotterdam, Netherlands, made of three identical square slabs of side 8 m. The three slabs are assembled perpendicular to each other such that they meet at one common point O . Each slab has one vertex fixed to the flat horizontal ground, represented by the plane π , and one vertex fixed to the centre of another slab.



- The slab p_A is part of the plane $x=0$. It has one vertex, point A , on π and another vertex on the centre of p_C .
- The slab p_B is part of the plane $y=0$. It has one vertex, point B , on π and another vertex on the centre of p_A . The remaining two vertices are denoted as D and E .
- The slab p_C is part of the plane $z=0$. It has one vertex, point C , on π and another vertex on the centre of p_B .

The point O is taken as the origin and perpendicular unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are defined, as shown in the diagram, with $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ along one edge of p_C , p_A , p_B respectively. For example, point D has position vector $8\mathbf{i} + 4\mathbf{k}$. It is assumed that the slabs are made of materials with negligible thickness.

(i) Show that a Cartesian equation of π is $x + y - z = 12$. [4]

(ii) To make the sculpture more stable, plans are made to construct a vertical support from O to a point F on π , such that OF is perpendicular to π . Find the position vector of F . [3]

A pigeon is at a point $P(2, 11, 1)$. It flies in a straight line and rests on the sculpture along the edge DE .

(iii) Given that the pigeon rests at the point nearest to P , find the coordinates of this point. Hence find the distance the pigeon flew. [4]

Another pigeon has flight path with equation $\frac{x-1}{3} = \frac{2-y}{4} = z+3$.

(iv) Without using a calculator, determine if the flight paths of the two pigeons cross. [3]